DAGMAWI TESFU TEMESGEN

Software Engineering Student | Full-Stack & AI Engineering

Addis Ababa, Ethiopia · GitHub: github.com/DagmTesfu · Portfolio: dagmawitesfu-portifolio.vercel.app

EDUCATION

Addis Ababa University — B.Sc. Computer Science

Expected Graduation: June 2028 · Addis Ababa, Ethiopia

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Advanced Database Systems, Computer Networks & Data Communications, Microprocessor & Assembly Language, Applied Numerical Analysis

EXPERIENCE

Frontend Development Intern / Contributor — Glynac / Sylvan-tech

Remote · 2026

- Contributed to frontend development and implementation of web application features in a remote software team.
- Built and refined responsive user interfaces with modern frontend technologies.
- Worked with existing codebases, translated requirements into functional UI, and collaborated through Git-based development workflows.
- Improved usability, responsiveness, and consistency across application interfaces.

Independent Software Engineering Projects

Remote · 2025–Present

- Designed and developed web applications for independent clients, translating business requirements into production-ready interfaces and functionality.
- Built responsive frontend experiences and integrated backend services, databases, APIs, and third-party services.
- Worked across the development lifecycle from requirements and architecture through implementation, testing, deployment, and iteration.
- Developed a web application for NTT Business Development, focusing on a practical business workflow and responsive user experience.
- Delivered a private client web application involving authentication, database-backed functionality, administrative features, and production deployment.

SELECTED PROJECTS

TabSplit — Receipt Splitting Platform

React · Vite · Node.js · Express · Supabase · PostgreSQL · AI Vision

- Built an Ethiopia-focused receipt-splitting application that lets users scan receipts, review extracted items, assign items to people, and calculate individual totals.
- Implemented server-side AI vision extraction with support for ETB and USD currencies.
- Built exact monetary calculations using integer minor-unit arithmetic and largest-remainder allocation to prevent floating-point errors.
- Designed a backend architecture where AI credentials and database access remain server-side.
- Added bill finalization, server-side validation and recalculation, shareable bill links, and generated SVG snapshots for finalized receipts.
- Created automated tests covering receipt-splitting and monetary calculation logic.

AccessPilot — Accessibility Engineering Platform

Next.js · TypeScript · Accessibility Testing · Docker · GitHub Integration · AI

- Built an accessibility engineering platform that scans web pages for WCAG 2.2 accessibility issues.
- Integrated automated accessibility analysis with contextual screenshots and source-code locations to make findings actionable.
- Developed AI-assisted explanations and grounded patch proposals using the affected source context.
- Implemented deterministic patch application and verification workflows to validate changes against the original source snapshot.
- Built a Docker-based execution environment and GitHub SHA-based source snapshot workflow for reproducible verification.
- Designed the system to keep developers in control by requiring approval before applying proposed fixes.

Zebo Cookies — Business Operations Platform

Node.js · Express · PostgreSQL · Supabase · React · Telegram Bot API

- Built a private business operations platform combining Telegram-based daily sales logging with a secure administrative financial approval portal.
- Developed a Telegram workflow allowing field representatives to record daily sales directly through the Telegram Bot API.
- Built administrative workflows for managers to review, reconcile, and approve daily collected-money submissions.
- Designed a custom algorithmic promotional lottery engine tailored to campaign-specific business rules.
- Implemented authentication and role-based access controls to protect proprietary operational and financial data.
- Designed PostgreSQL-backed data models and REST APIs connecting sales workflows, administrative tools, and business records.

- Built and integrated a React administrative interface with server-side validation and controlled approval workflows.
- Project is covered by an NDA; client-specific data, screenshots, and proprietary business logic are not publicly disclosed.

GroupMemory — Searchable Telegram Group Archive

React · JavaScript · Node.js · Express · Supabase · PostgreSQL · Telegram Bot API

- Built a Telegram-based group archive that captures and organizes group messages for later search and retrieval.
- Designed a relational PostgreSQL data model for groups, senders, and messages with Telegram identifiers and uniqueness constraints.
- Implemented Telegram Bot API polling and backend health monitoring.
- Built the application with a React frontend and Express backend connected to Supabase.
- Focused on reliable message ingestion and structured storage rather than adding unnecessary complexity.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, SQL, HTML, CSS

Frontend: React, Next.js, Vite, Tailwind CSS, Responsive UI, Accessibility

Backend: Node.js, Express.js, REST APIs, API Integration, Authentication

Databases: PostgreSQL, Supabase, Database Design, SQL

AI Engineering: AI API Integration, Vision/Multimodal APIs, Structured AI Outputs, OpenRouter

Testing & Quality: Automated Testing, API Testing, Accessibility Testing, Input Validation

DevOps & Tools: Git, GitHub, Docker, Vercel, Linux/CLI, Environment Configuration

Engineering: Full-Stack Development, Software Architecture, API Design, Debugging, Deployment

ADDITIONAL

- Strong interest in Software Engineering, AI Engineering, accessibility, and developer tooling.
- Experienced building applications from idea → architecture → implementation → testing → deployment.
- Comfortable working independently and contributing within remote development teams.